JOSSE XGBoost: Dataset and Three Experiments

1. Dataset Information

Dataset: Original JOSSE archive, source.zip/source/JOSSE_18092020.sqlite3 (SQLite table case).

Original JOSSE archive: 23,186 rows and 7 columns.

Our dataset preparation criteria included:

1. **Valid effort:** Tasks needed recorded working hours between 0.1 and 240 hours.
2. **Sufficient text:** Task descriptions needed at least 10 characters.
3. **Valid identifiers:** Tasks needed valid issue and project identifiers.
4. **No duplicates:** Duplicate issue IDs and identical normalized task texts were excluded.

Filtered experimental dataset: 15,341 tasks across **335 projects**.

Columns in the original SQLite table

Column	Meaning	Model role
id	Issue ID (such as ACCUMULO-1010)	Identifier / join; not a feature
corpus	Issue title and description text	Only source feature
num_comment	Number of comments	Excluded
num_activities	Recorded activity count	Excluded
expert_estimated_effort	Archived estimate (many -1 placeholders)	Excluded; timing not verified
actual_effort	Recorded effort in seconds	Target source (divide by 3,600)

2. Training Details

Algorithm: XGBoost (gradient-boosted decision trees) in all three versions.

- **V1:** Predicts the number of working hours a task requires.
- **V2:** Still predicts working hours, but we tried improving its accuracy.
- **V3:** Predicts working hours AND trains two additional classifiers:
    - **Long-task classifier:** Predicts whether a task will take more than 24 hours.
    - **Short-task classifier:** Predicts whether a task will take 0.5 hours or less.

The training labels are recorded hours; labels and activity counts are never used as prediction inputs.
Preprocessing: TF-IDF converts words in issue text into weighted numbers. SVD compresses those numbers into fewer features.

Key verified configurations

Version	Input representation	Important model settings
V1 baseline	64 text SVD dimensions	XGBRegressor, squared-error; 250 trees, depth 5, learning rate 0.04.
V2 retrained robust	64 SVD + 21 text-derived statistics = 85	Pseudo-Huber loss; 240 trees, depth 3, learning rate 0.04. V2 also tests 3 other regressors.
V3 trained	64 word SVD + 24 title-character SVD + 21 text statistics = 109	Squared-error regressor; 320 trees, depth 3, learning rate 0.05. Long/short classifiers: 300/200 trees.

Frozen project-disjoint partitions

Partition	Tasks	Share	Projects	Used for
TRAIN	10,717	69.86%	233	Fit models and text transformations
VALIDATION	2,324	15.15%	55	Compare models and inspect errors
FINAL_TEST_LOCKED	2,300	14.99%	47	Not evaluated

Projects do not overlap across the three partitions.

How the validation metrics work

Metric	Meaning	Better
MAE (hours)	Average absolute difference between actual and predicted working hours	Lower
PRED(25)	Share of predictions within 25% of the actual hours	Higher
MdMRE	Median absolute percentage error relative to actual hours	Lower
Numeric >24h recall	Among actual tasks >24h, share predicted numerically >24h	Higher

3. Experiment 1 - Baseline

What was done: Trained a text-only TF-IDF/SVD + XGBoost regressor and recorded an 11-configuration search. The saved baseline uses the V1 settings above

V1 recorded result (VALIDATION)	MAE ↓	PRED(25) ↑	MdMRE ↓	>24h
Saved baseline	5.8418h	9.036%	92.076%	0/139
Trial 10, recorded score; weights not saved	5.8386h	10.026%	92.193%	0/139

Setting	V1 Baseline	Trial 10
Algorithm	XGBoost	XGBoost
Input features	64 SVD text features	64 SVD + 21 text statistics
Loss function	Squared Error	**Pseudo-Huber**
Number of trees	250	240
Maximum tree depth	5	3
Learning rate	0.04	0.04 (unchanged)
Minimum child weight	1	10

Problems:

- Very low PRED(25),
- large errors on longer tasks
- no numerical prediction above the 24h threshold among 139 actual long tasks
- Selection discrepancy: the report scores Trial 10 better in MAE/PRED(25)

4. Experiment 2 - First Improvement (V2)

Main change - calibration, not retraining: scripts/recalibrate.py:main() multiplies the existing baseline hour predictions by a constant. It tested factors 0.50 through 1.00 in steps of 0.05, choosing the lowest validation MAE without reducing baseline PRED(25). The selected factor was 0.70. Model weights, input text, and original split stayed unchanged.

V2 calibration result

Validation variant	MAE ↓	PRED(25) ↑	MdMRE ↓	Numeric >24h
V2 calibrated (x0.70)	5.7983h	10.370%	88.861%	0/139

Compared with the saved V1 baseline, MAE fell by 0.0435h (about 2.6 minutes), PRED(25) increased by 1.334 percentage points, and MdMRE fell by 3.215 points. This was a post-training correction; it did not teach XGBoost new patterns. The factor was selected on reused validation.

V2 also retrained new XGBoost models

retraining/run_josse_next_experiment.py:main() trained four new candidates using the fixed 10,717 TRAIN tasks, 64-dimensional word SVD, and optional 21 text statistics. These results are distinct from the x0.70 adjustment.

V2 candidate (VALIDATION)	MAE ↓	PRED(25) ↑	MdMRE ↓
New log-hours baseline	5.8432h	9.466%	92.424%
Robust pseudo-Huber retraining	5.8295h	9.208%	92.152%
Long-task-weighted log-hours	6.2282h	9.897%	161.823%
Direct-hours absolute-error	5.8431h	8.735%	92.608%

The robust retrained model was selected relative to the original uncalibrated baseline, but it did not beat V2 calibration on MAE or PRED(25). The weighted run worsened MAE markedly. All four new numeric models still detected 0/139 actual tasks above 24h.

Separate risk score and exploratory adjustment

A new >24h risk classifier was trained (not an hours regressor). Flagging the top 20% risk scores caught 57/139 long tasks (41.01% recall; 12.26% precision). Later, JOSSE_Targeted_Validation_Analysis/reproduce_validation.py adjusted saved hour predictions using those scores without retraining: MAE 5.7867h, PRED(25) 10.714%, MdMRE 88.797%. This result is exploratory because the same validation data were repeatedly examined. Numeric long-task recall remained 0/139.

Evidence: scripts/recalibrate.py; results/v2_generated/V2_CALIBRATION_REPORT.json; optional_retraining.zip/NEW_EXPERIMENT_REPORT.json and NEW_VALIDATION_TRIALS.csv; JOSSE_Targeted_Validation_Analysis/COMPARISON.csv.

5. Experiment 3 - Second Improvement (V3)

Main change - real retraining: research/07_train_risk_aware_v3.py adds title-character patterns and text statistics to the word features, creating 109 numeric text features. It retrains a new hour XGBoost regressor plus two classifiers: long-task risk (>24h) and short-task likelihood (≤0.5h). It uses the same frozen TRAIN and VALIDATION partitions.

Prediction adjustment: Three project-group training folds produced out-of-fold predictions for choosing the V3 rule. Selected without validation labels: scale=0.65, risk_gain=10, risk_start=0.60; no short-task gate. Separately, V3 explored combinations using the earlier V2 calibrated predictions and a short-task adjustment. These post-hoc combinations were examined on reused validation; they are not independently verified gains.

V3 validation results

V3 variant	MAE ↓	PRED(25) ↑	MdMRE ↓	Numeric >24h
New raw regressor	5.8483h	8.907%	91.998%	0/139
Train-only selected rule	5.8017h	9.897%	88.621%	0/139
Risk + V2 calibrated (exploratory)	5.7950h	10.370%	88.748%	0/139
Short/risk + V2 (post-hoc exploratory)	5.8073h	14.673%	87.674%	0/139

The train-only selected V3 rule is slightly worse than V2 calibration on MAE (5.8017h vs 5.7983h) and PRED(25) (9.897% vs 10.370%), but slightly better on MdMRE. The post-hoc short/risk combination has higher PRED(25), at the cost of worse MAE than V2 calibration.

Long tasks: As an independent warning output, V3 flags the highest-risk 20% (465 tasks): 59 of 139 actual tasks >24h are caught (42.45% recall; 12.69% precision; 406 false alarms). None of the numeric hour predictions correctly exceeds 24h on those 139 tasks.

Final comparison - same 2,324-task VALIDATION set

Version / recorded variant	Retrained?	MAE ↓	PRED(25) ↑	MdMRE ↓
V1 saved baseline	Yes	5.8418h	9.036%	92.076%
V1 Trial 10 (weights unavailable)	Reported trial	5.8386h	10.026%	92.193%
V2 calibrated (reference)	No	5.7983h	10.370%	88.861%
V2 robust refit	Yes	5.8295h	9.208%	92.152%
V2 risk-adjusted (exploratory)	No	5.7867h*	10.714%	88.797%
V3 train-only selected	Yes + rule	5.8017h	9.897%	88.621%
V3 short/risk + V2 (exploratory)	V3 classifiers	5.8073h	14.673%*	87.674%*

* Best displayed XGBoost-based result on that metric, but exploratory and not independently validated. V2 calibrated remains the clearest overall hour-estimation reference. V1/V2/V3 all used the same repeatedly reviewed validation tasks; the locked 2,300-task final test was not scored. No V4 is present. Evidence: JOSSE_V3_Results.zip/V3_REPORT.json; V3_VALIDATION_PREDICTIONS.csv; research/07_train_risk_aware_v3.py.